## Twisted line on the Oakley EC2N curves

Run run-20261027T1445Z, seed 0xAC120261027, 200 samples per row. Source of every number: `reports/run-20261027T1445Z/rows.json`.

| curve | N | d | n | Tr(a) | u found (lines) | twisted lift | standard lift |
|---|---|---|---|---|---|---|---|
| Oakley 3 | 155 | 31 | 5 | 0 | yes (6) | 200/200 | 104/200 |
| Oakley 4 | 185 | 37 | 5 | 0 | yes (3) | 200/200 | 99/200 |
| c2pnb176v1 | 176 | 44 | 4 | 1 | yes (3) | 0/200 | 122/200 |
| c2pnb208w1 | 208 | 52 | 4 | 0 | yes (1) | 200/200 | 108/200 |
| c2pnb272w1 | 272 | 68 | 4 | 1 | yes (2) | 0/200 | 103/200 |
| c2pnb304w1 | 304 | 76 | 4 | 1 | yes (5) | 0/200 | 98/200 |
| c2pnb368w1 | 368 | 92 | 4 | 0 | yes (3) | 200/200 | 109/200 |

Positive control behaves as predicted. On both Oakley curves a u with T(u) = T(c/u) = 0 exists and was verified directly in $F_{2^N}$; all sampled elements of the line $uF_q$ lift. If the twisted line is used as the Gaudry-Diem base for n = 5, each nonzero x lifts, so a decomposition trial has 2^5 = 32 times the relations of the standard base. This is a per-trial relation-yield statement, not a measured cost; composition with the planned subfield-base lane and the GHS genus-16 transfer pricing are not examined.

| search to first u (s) | Oakley 3 | Oakley 4 |
|---|---|---|
| | 0.046 | 0.051 |

Table 1: Timing on a 4-core Xeon 2.10 GHz VM; whole run 64.5 s.